

Contextual Embeddings Can Distinguish Homonymy from Polysemy in a Human-Like Way

Kyra Wilson¹, Alec Marantz^{1 2}

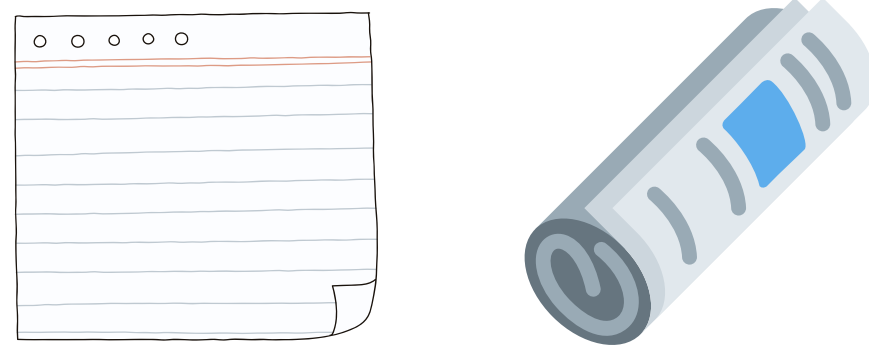
¹New York University Abu Dhabi, ²New York University ✉ kew6086@nyu.edu

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INTRODUCTION

POLYSEMY

'paper'



HOMOPHONY

'bark'



- Language models typically do not distinguish between types of ambiguity, and recent models use contextualized word embeddings to vary the representation of words based on their surrounding contexts
- Psycholinguistic studies show human processing works differently. Polysemous words are processed faster than unambiguous words, and homophonous words are either slower to recognize or the same as unambiguous words
- Part of the difficulty in studying ambiguous word processing in humans is that measures of ambiguity rely on sources which have been created by humans. These can be subject to bias and difficult to reproduce between studies

OBJECTIVES: Can contextualized word embeddings be used to quantify ambiguity? Do they perform as well as previous measures in predicting human behavior?

AMBIGUITY MEASURES

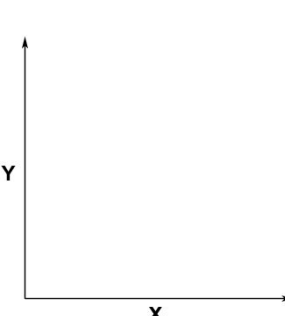
NUMBER OF SENSES



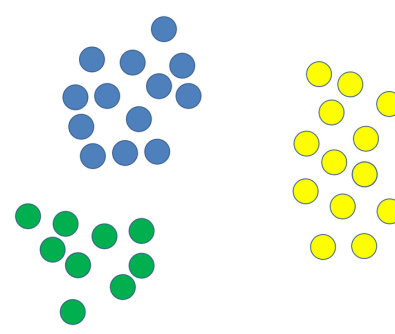
Sample 1,000 occurrences of each word from English Wikipedia following approach in [1]



Use BERT_{BASE} [2] to get contextualized embedding of each word in its sentence-length context



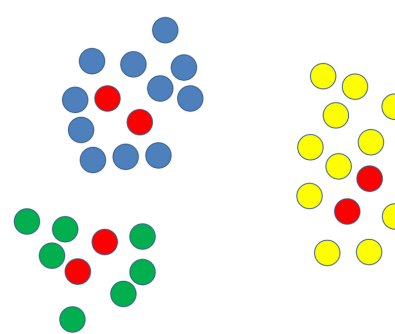
Apply tSNE to reduce dimensions from 758 to 2. This helps with visualization of embeddings and accuracy of clustering



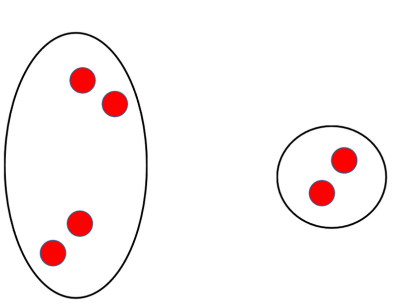
Use HDBSCAN [3] to cluster the embeddings for a given word with a minimum cluster size of 10 embeddings. The algorithm does not specify a number of clusters a priori, and points may be identified as noise

**NUMBER OF CLUSTERS =
NUMBER OF SENSES**

NUMBER OF MEANINGS



Using HDBSCAN identify exemplar points in each of the sense clusters, discarding outliers and noise



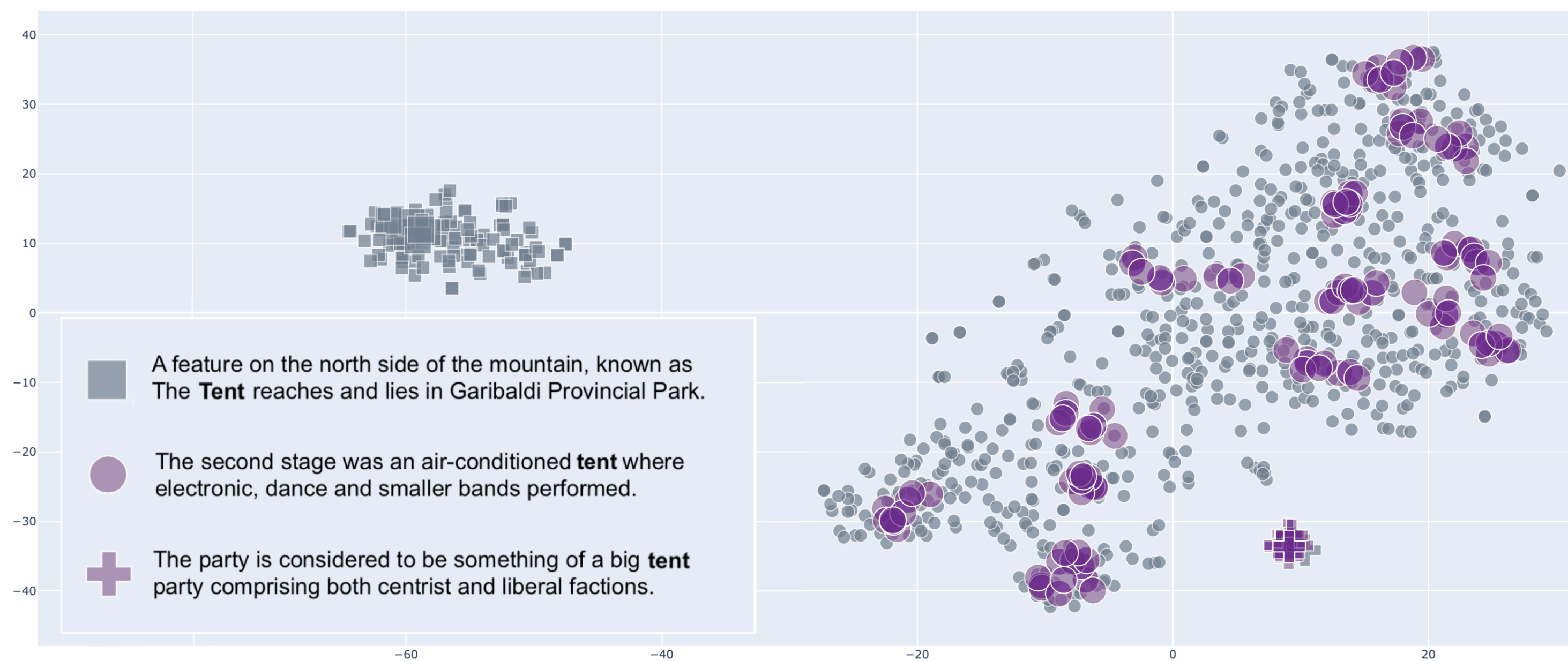
Apply the clustering algorithm to just the exemplar points with the minimum cluster size equal to the number of exemplar points in the smallest sense cluster

**NUMBER OF SECOND-ORDER
CLUSTERS = NUMBER OF MEANINGS**

REFERENCES

[1] Emily Reif, Ann Yuan, Martin Wattenberg, Fernanda B Viegas, Andy Coenen, Adam Pearce, and Been Kim. 2019. Visualizing and measuring the geometry of BERT. *Advances in Neural Information Processing Systems*, 32.
[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1 (Long and Short Papers), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
[3] Ricardo JGB Campello, Davoud Moulavi, and Jörg Sander. 2013. Density-based clustering based on hierarchical density estimates. In *Pacific-Asia conference on knowledge discovery and data mining*, pages 160–172. Springer.
[4] Jennifer Rodd, Gareth Gaskell, and William Marslen-Wilson. 2002. Making sense of semantic ambiguity: Semantic competition in lexical access. *Journal of Memory and Language*, 46(2):245–266.
[5] Balota, D.A., Yap, M.J., Cortese, M.J., Hutchison, K.A., Kessler, B., Loftis, B., Neely, J.H., Nelson, D.L., Simpson, G.B., & Treiman, R. (2007). The English Lexicon Project. *Behavior Research Methods*, 39, 445-459.

EXAMPLE CLUSTERING FOR 'TENT'



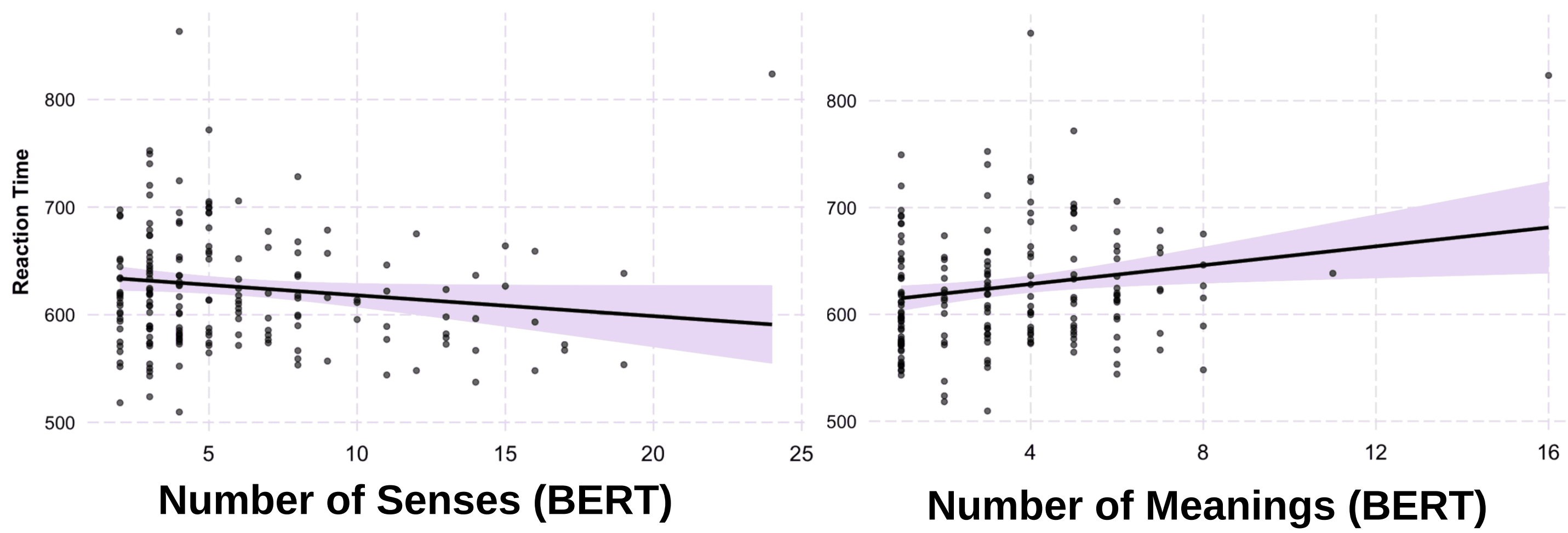
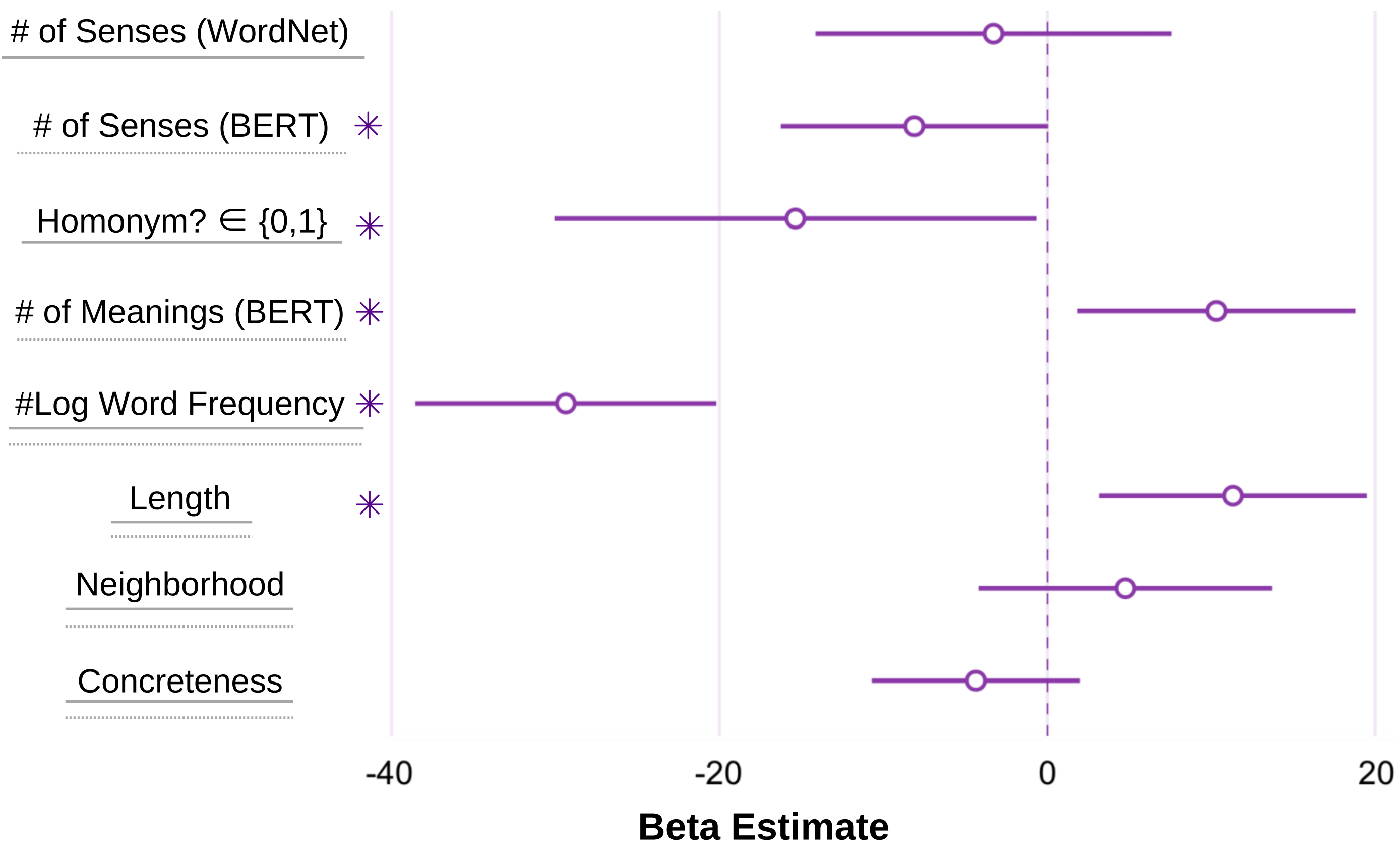
Different senses are indicated by different shapes, exemplar points used to cluster meanings are larger, and purple points indicate meaning groups.

EXPERIMENT AND RESULTS

Analysis: Compare our measures with those used in [5] in separate linear regressions

Data: 182 words (124 ambiguous, 58 unambiguous) from [4] and reaction times from English Lexicon Project [5]

$$RT \sim \# \text{ Senses} + \# \text{ Meanings} + \log \text{ Word Frequency} + \text{Length} + \text{Neighborhood} + \text{Concreteness}$$

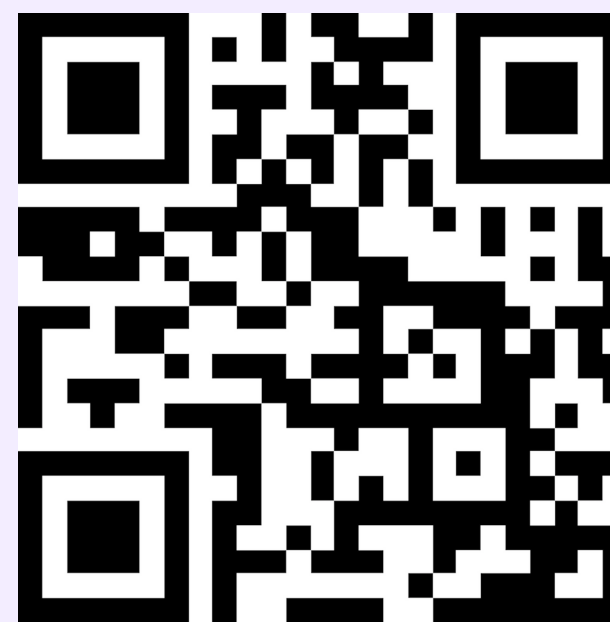


Additional ANOVA tests showed model fit was improved by both senses ($F=3.78$, $p=0.05$) and meanings ($F=6.61$, $p=0.01$)

CONCLUSION

- It is possible to **use contextual embeddings to derive measures of homonymy and polysemy**
- **These measures correspond to human behavior** in single-word lexical decision tasks
- Using CWEs to derive measures of ambiguity will improve psycholinguistic research because they are **easier to replicate across experiments, less prone to experimenter bias, and easier to extend to languages other than English**

Poster PDF
and Code



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